CONSTRUCTION PROGRAMMES

CURRICULUM CHART FOR

TRADE TEST LEVEL II CERTIFICATE IN CARPENTRY AND JOINERY

CHART NO. 70B2

Prepared by:

Curriculum Development Unit

Development Division

TEVETA

November, 2007

TABLE OF CONTENTS

ITEM PAGE

1.0 ACKNOWLEDGEMENTS I
2.0 RATIONALE III
3.0 PROGRAMME PURPOSE III
4.0 PROGRAMME OBJECTIVES III
5.0 COURSE DURATION IV
6.0 COURSE OUTLINE IV
7.0 TEACHING STRATEGIES IV
8.0 PROGRAMME EVALUATION IV
9.0 ENTRY REQUIREMENTS V
10.0 ASSESSMENT VI
11.0 STAFFING VI
12.0 CERTIFICATION VI
MODULE 70B2-01-A HEALTH AND SAFETY 1
MODULE 70B2-02-A WOOD WORKING HAND TOOLS AND MACHINERY 2
MODULE 70B2-03-A WOOD WORKING MATERIALS 3
MODULE 70B2-04-A CONSTRUCTION OF BUILDING UNITS AND FITTINGS ... 4
MODULE 70B2-05-A TRADE DRAWING 6
MODULE 70B2-06-A TRADE MATHEMATICS 7
MODULE 70B2-07-A COMMUNICATION SKILLS 8
MODULE 70B2-08-A COMPUTER APPLICATIONS 9
MODULE 70B2-09-A ENTREPRENEURSHIP 10
MODULE 70B2-10-A ROOF CONSTRUCTION 12
MODULE 70B2-11-A TEMPORAL TIMBER STRUCTURES 14

1.0 ACKNOWLEDGEMENTS

The Technical Educational, Vocational and Entrepreneurship Training Authority (TEVETA) takes this opportunity to express sincere appreciation to the Carpentry and Joinery Programmes Curriculum Development and Review Team for the support in the production of this document.

The curriculum has been reviewed with the assistance of the Team comprising the following people:

	Name	Institution
1.	Moses Tembo	Dzithandizeni Trades School
2.	Boaz Daka	Lusaka Business and Technical College
3.	Wilson Kabonga	St Ambrose Trades
4.	Francis Mwape	National Council for Construction
5.	James Banda	M&K Enterprise
6.	Sam Phiri	Apollo
7.	Patson Chipili	Entrepreneur
8.	Dickson R Ngosa	Community Based
9.	Stellah Mungaila	TEVETA
10.	Hazel Njoloma	TEVETA
11.	Catherine M. Kanyensha	TEVETA

TEVETA also wishes to extend acknowledgement to the following secretary who typed both the chart and syllabus of this curriculum.

1.	Ms Maboshe Inambao	UNZA

The validation team comprised the following:-

1.	Dr M.S. Mashamba	National Council for Construction
2.	Mr Medson Simumba	Lusaka Business and Technical College
3.	Mr Boaz Daka	Thorn Park Construction Training Centre

4.	Mr David Chaponda	Rockwell Engineering
5.	Mr Boniface Sambwa	Lusaka Vocational Training Centre
6.	Mr Chitengi Kahalu	Lusaka Business and Technical College
7.	Mr Nyonge Phiri	Lusaka Water and Sewerage Company
8.	Mr Costain Silweya	Examinations Council of Zambia
11.	Catherine M. Kanyensha	TEVETA

The Secretarial services were offered by

1. Mrs Annette M. Mwamulima

2.0 RATIONALE

The social, economic, cultural, demographic and political changes that have occurred in Zambia have not spared the construction industry which is one of the oldest in the country.

Liberalization of the economy and privatization of parastatal companies in the early 1990s saw a decline in standards in the construction industry. It is against this background that the Trade Test Level II Certificate in Carpentry and Joinery has been reviewed to meet the current needs of the construction industry. The country therefore, needs qualified human resource that can competently and efficiently perform carpentry and joinery works, manage the site/workshop and help revamp the construction industry.

This curriculum is aimed at providing the nation with competent persons that are able to perform carpentry and joinery works as handymen.

3.0 PROGRAMME PURPOSE

To provide both formal and informal sectors of the economy with competent handy persons, able to effectively utilise skills, knowledge and attitudes in competently carrying out various carpentry and joinery tasks.

4.0 PROGRAMME OBJECTIVES

On completion of this programme, trainees will be able to:

3 Observe health and safety measures
4 Produce carpentry and joinery work products
5 Construct building units and fittings
6 Erect temporal timer structures and gabled roofs
7 Use word processing packages
8 Apply entrepreneurial skills
9 Communicate effectively and efficiently

5.0 COURSE DURATION

6 months

6.0 COURSE OUTLINE

MODULE NO.	MODULE TITLE	NO. OF HRS
70B2-01-A	HEALTH AND SAFETY	40
70B2-02-A	WOOD WORKING HAND TOOLS AND MACHINERY	50
70B2-03-A	WOOD WORKING MATERIALS	120
70B2-04-A	CONSTRUCTION OF BUILDING UNITS AND FITTINGS	320
70B2-05-A	TRADE DRAWING	40
70B2-06-A	TRADE CORE MATHEMATICS	40
70B2-07-A	COMMUNICATION SKILLS	60
70B2-08-A	COMPUTER APPLICATIONS	60
70B2-09-A	ENTREPRENEURSHIP	80
70B2-10-A	ROOF CONSTRUCTION	100
70B2-11-A	TEMPORAL TIMBER STRUCTURES	40
	TOTAL	**950**

7.0 TEACHING STRATEGIES

The teaching methods will include lectures, practical demonstrations, practical project and applied management routine activities, construction site tours and guided reading.

8.0 PROGRAMME EVALUATION

TEVETA shall evaluate the programme as follows:

8.1 FORMATIVE EVALUATION

This will be done in order to determine, on an ongoing basis, whether the programme is being implemented as planned and to advise on improvements.

8.1.1 Major areas of evaluation

Course aims and objectives, trainees' entry requirements, course content, teaching /learning activities and resources.

8.1.2 Evaluation instruments

Questionnaires, structured interviews, observations, checklists, examination results.

8.1.3 Sources of information

Trainees, teachers, administrators, college dropouts and field instructors.

8.2 SUMMATIVE EVALUATION

This is done at the end of a program to determine whether the goals are being achieved.

8.2.1 Major areas of evaluation

Course aims, trainee entry requirements, course content, learning resources, teaching / learning activities, teachers and graduates performance in employment.

8.2.2 Evaluation instruments

Questionnaires, structured interviews, observations, projects, checklists, Records of work and examination results.

8.2.3 Sources of information

Information for the evaluation will be obtained from trainees, teachers, administrators, college dropouts, employers, examination results, union officials and the general public.

9.0 ENTRY REQUIREMENTS

9.1 Direct Entry: Grade Nine (9) level of education.

9.2 **Mature Age:** Grade 7 Certificate with Level III Trade Test Certificate in Carpentry and Joinery and one year relevant industrial experience.

10.0 ASSESSMENT

Final Trade Test	**100%**
Minimum pass Mark	**70%**

11.0 STAFFING

11.1 Lecturing staff must have at least a Craft Certificate in Carpentry and Joinery or a related carpentry and joinery qualification with not less than 1 year relevant work experience

11.2 Level 1 Trade Test Certificate in carpentry and Joinery with three years relevant work experience and a teaching qualification

12.0 CERTIFICATION

Trade Test Level II in Carpentry and Joinery Certificate will be awarded by the Examinations Council of Zambia (ECZ).

MODULE 70B2-01-A HEALTH AND SAFETY

NOMINAL DURATION: 40 HOURS 4 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes to enable them observe health and safety regulations in the construction industry.

OBJECTIVES: On completion of this module, trainees will be able to:

3 Observe hand tools and machine safety regulations

4 Practice personal safety

5 Observe workshop safety

UNIT A1.1 OBSERVING HAND TOOL AND MACHINE SAFETY

1.1.1 Condition and use

1.1.2 Government regulations

1.1.3 Machine guards

UNIT A1.2 PRACTICING PERSONAL SAFETY

1.2.1 Eye and skin protection

1.2.2 Clothing

1.2.3 Behaviour

1.2.4 Treatment of injuries

1.2.5 Artificial respiration

1.2.6 HIV and AIDS

UNITA1.3 OBSERVING WORKSHOP SAFETY

1.3.1 House keeping

1.3.2 Fire hazards and protection

1.3.3 Warning signs and danger tags

MODULE 70B2-02-A WOOD WORKING HAND TOOLS AND MACHINERY

NOMINAL DURATION: 50 HOURS 5 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes in wood working hand tools and machinery to be applied in the construction industry.

OBJECTIVES: On completion of this module, trainees will be able to:

1. Use hand tools
2. Operate simple portable and pneumatic tools
3. Operate simple wood working machines

UNIT A2.1 USING HAND TOOLS

2.1.1 Types of hand tools
2.1.2 Uses of hand tools
2.1.3 Maintenance of hand tools
2.1.4 Storage of hand tools

UNIT A2.2 OPERATING SIMPLE PORTABLE AND PHEUMATIC TOOLS

2.2.2 Types of portable and pneumatic tools
2.2.3 Uses of portable and pneumatic tools
2.2.3 Maintenance of portable and pneumatic tools
2.2.4 Storage of portable and pneumatic tools

UNIT A2.3 OPERATING SIMPLE WOOD WORKING MACHINES

2.3.1 Types of wood working machines
2.3.2 Setting machines
2.3.3 Uses of wood working machines
2.3.4 Maintaining wood working machines

MODULE 70B2-03-A WOOD WORKING MATERIALS

NOMINAL DURATION: 120 HOURS 12 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes in wood working materials to be applied in the construction industry.

OBJECTIVES: On completion of this module, trainees will be able to:

3 Use timber

4 Use manufactured boards

5 Use hardware

UNIT A3.1 USING TIMBER

3.1.1 Classification of timber

3.1.2 Structure of timber and properties

3.1.3 Convention of timber

3.1.4 Seasoning of timber

3.1.5 Timber defects

3.1.6 Marketing sizes of timber

3.1.7 Preservation of timber

3.1.8 Wood finishes

3.1.9 Storage of timber

UNIT A3.2 USING MANUFACTURED BOARDS

3.2.1 Types of manufactured boards

3.2.2 Manufacturing process

3.2.3 Application of manufactured boards

3.2.4 Storage of manufactured boards

UNIT A3.3 USING HARDWARE

3.3.1 Types of ironmongery

3.3.2 Uses of hardware

3.3.3 Types of adhesives

MODULE 70B2-04-A CONSTRUCTION OF BUILDING UNITS AND FITTINGS

NOMINAL DURATION: 320 HOURS 32 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes to enable them construct and erect building units and fittings to satisfy the construction industry and customers.

OBJECTIVES: On completion of this module, trainees will be able to:

1. Make mortise and tenon joints
2. Construct workshop tools
3. Construct small casement windows
4. Construct small cabinet with framed door

UNIT A4.1 MAKING MORTICE AND TENON JOINTS

4.1.1 Types of mortise and tenon joints

4.1.2 Prepare required material

4.1.3 Make various mortise and tenon joints

UNIT A4.2 CONSTRUCTING WORKSHOP TOOLS

4.2.1 Interpreting drawing

4.2.2 Preparation of materials

4.2.3 Working out positions of rails on legs

4.2.4 Cutting out mortises by hand

4.2.5 Cutting all tenons by hand

4.2.6 Finish all members before assembly

4.2.7 Gluing on of frame

4.2.8 Fixing top

4.2.9 Finishing with appropriate finish

UNIT A4.3 CONSTRUCT SMALL CASEMENT WINDOW WITH FRAME

4.3.1 Preparation of materials

4.3.2 Preparation of setting out plan on sheet of particle board

4.3.3 Marking out of members

4.3.4 Mortising members

4.3.5 Tenoning rails

4.3.6 Beveling and rebating

4.3.7 Gluing frame and sash

4.3.8 Fitting of sash into frame

4.3.9 Finishing with appropriate paint

UNIT A4.4 CONSTRUCT SMALL CABINET WITH FRAMED DOOR

4.4.1 Preparation of materials

4.4.2 Marking position of top/bottom on sides

4.4.3 Sanding and finishing all members before assembling car case

4.4.4 Assembling car case

4.4.5 Fitting in plywood back

4.4.6 Fixing of door with butt hinges

4.4.7 Fixing of magnetic catch

4.4.8 Finishing with sealer and clear varnish

MODULE 70B2-05-A TRADE DRAWING

NOMINAL DURATION: 40 HOURS 4CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes to enable them apply trade drawing related to carpentry and joinery to the satisfaction of the construction industry and customers.

OBJECTIVE:**On completion of this module, trainees will be able to**:

1. Handle drawing instruments
2. Draw elementary drawings
3. Produce elementary scaled drawings

UNIT A5.1 HANDLING DRAWING INSTRUMENTS

5.1.1 Types of drawing instruments

5.1.2 Uses of drawing instruments

5.1.3 Care of drawing instruments

UNIT A5.2 DRAWING PLANS AND ELEVATIONS

5.2.1 Types of projections

5.2.2 Polygons

5.2.3 Arch centres

5.2.4 Angles

UNIT A5.3 PRODUCING SCALED DRAWINGS

5.3.1 Measurements

5.3.2 Convention

5.3.3 Reading dimensions

5.3.4 Symbols used in drawings

MODULE 70B2-06-A TRADE MATHEMATICS

NOMINAL DURATION: 40 HOURS 4 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes to enable them apply basic mathematics in the construction industry.

OBJECTIVES: On completion of this module, trainees will be able to:

1. Carry out basic calculations
2. Calculate length, area and volume

UNIT A6.1 CARRY OUT BASIC CALCULATIONS

6.1.1 Ratios and proportions quantities

6.1.2 Metric measures

6.1.3 Calculate percentages

6.1.4 Average

6.1.5 Quantities and decimal places

UNIT A6.2 CALCULATING LENGTH, AREA AND VOLUME

6.2.1 Areas of familiar shapes

6.2.2 Measure length, angle and weight

MODULE 70B2-07-A COMMUNICATION SKILLS

NOMINAL DURATION: 60 HRS 6 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes to enable them communicate effectively in the construction industry.

OBJECTIVES: On completion of this module, trainees will be able to:

1. Explain the process of effective communication
2. Apply oral communication
3. Apply job seeking skills
4. Communicate effectively with others in the organization

UNIT A7.1 EXPLAINING THE PROCESS OF EFFECTIVE COMMUNICATION

7.1.1 Nature and purposes of communication

7.1.2 The communication process

7.1.3 Barriers of effective communication

7.1.4 Strategies of effective communication

UNIT A7.2 APPLYING ORAL COMMUNICATION

7.2.1 Listening
7.2.3 Note taking / Note making
7.2.1 Non-verbal communication

UNIT A7.3 APPLYING JOB SEEKING SKILLS

7.3.1 Writing application letters
7.3.2 Types of interviews
7.3.3 Planning for and attending interviews
7.3.4 Interview procedures

UNIT A7.4 COMMUNICATING EFFECTIVELY IN ORGANISATIONS

7.4.1 Consultation with colleagues, supervisors and subordinates

7.4.2 Carrying out joint consideration of problems

7.4.3 Giving and receiving instructions

7.4.4 Seeking permission

7.4.5 Dealing with customers

7.4.6 Report writing

MODULE 70B2-08-A COMPUTER APPLICATIONS

NOMINAL DURATION: 60 HOURS 6 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes in the use of computer application packages in the construction industry.

OBJECTIVES: On completion of the course, the trainees will be able to:

1. Use windows operating systems
2. Use word processing packages e.g. Microsoft word.

UNIT A8.1 USING WINDOWS OPERATING SYSTEMS

8.1.1 Windows desktop
8.1.2 Starting a computer
8.1.3 Shutting down a computer
8.1.4 Opening Windows Explorer window
8.1.5 The parts of a window
8.1.6 Creating a new folder in window Explorer
8.1.7 Manipulating files and folders
8.1.8 Selecting a screen saver
8.1.9 The parts of a window
8.1.10 Switching between windows

UNIT A8.2 USING WORD PROCESSING PACKAGES - MICROSOFT WORD

8.2.1 The various parts of the MS – Word Screen
8.2.2 Creating a simple document
8.2.3 Simple editing tools
8.2.4 Saving the document and existing
8.2.5 Retrieving the document
8.2.6 Formatting the document
8.2.7 Using windows menu
8.2.8 Creating tables
8.2.9 Working with multiple documents
8.2.10 Getting help from Ms-Word

MODULE 70B2-09-A ENTREPRENEURSHIP

NOMINAL DURATION: 80 HOURS 8 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes in entrepreneurship to enable them apply these in the construction industry.

OBJECTIVES: On completion of this module, trainees will be able to:

1. Apply entrepreneurial concepts in carpentry and joinery
2. Develop entrepreneurial competences and attitudes
3. Apply enterprise management skills
4. Establish a carpentry and joinery business entity
5. Sustain a carpentry and joinery business

UNIT A9.1 APPLYING ENTREPRENEURIAL CONCEPTS

9.1.1 Concepts of Entrepreneurship

9.1.2 Economic trends in the carpentry and joinery industry in Zambia

9.1.3 Government policy on carpentry and joinery enterprise development

UNIT A9.2 DEVELOPING ENTREPRENEURIAL COMPETENCES AND ATTITUDES

9.2.1 Developing self-motivation

9.2.2 Identifying business opportunities

9.2.3 Networking for enterprise development

9.2.4 Effective business communication

UNIT A9.3 ESTABLISHING CARPENTRY AND JOINERY BUSINESS

9.3.1 Identify Business opportunities

9.3.2 Marketing

9.3.3 Cost and price per product / service

9.3.4 Management of finances

9.3.5 Business records

9.3.6 Business ethics and values in managing a carpentry and joinery business

9.3.7 Effective business communication

UNIT A9.4 ESTABLISHING A CARPENTRY AND JOINERY ENTERPRISE

9.4.1 Business Plan / project

9.4.2 Procedures for formalising an enterprise

MODULE 70B2-10-A ROOF CONSTRUCTION

NOMINAL DURATION: 100 HOURS 10 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes to enable them construct and erect roof trusses and gabled roofs to the satisfaction of the construction industry and customers.

OBJECTIVES: On completion of this module, trainees will be able to:

1. Interpret drawings of a roof truss
2. Construct and erect a roof truss
3. Interpret plans for a gabled roof
4. Construct and erect a gabled roof
5. construct and erect a flat roof

UNIT A10.1 INTERPRETING DRAWINGS OF ROOF TRUSSES

10.1.1.1 Types of each roof truss

10.1.1.2 Developing material list for roof truss

10.1.1.3 Joints used

UNIT A10.2 CONSTRUCTING AND ERECTING A ROOF TRUSS

10.2.1.1 Selecting suitable materials

10.2.1.2 Laying out timber and cutting members

10.2.1.3 Selecting connectors and assembling trusses

10.2.1.4 Erecting and supporting trusses

10.2.1.5 Applying wood preservatives.

UNIT A10.3 INTERPRETING PLANS FOR A GABLED ROOF

10.3.1.1 Reading drawing to determine construction requirements

10.3.1.2 Defining framing terms

10.3.1.3 Developing material list for a gabbled roof

UNIT A10.4 CONSTRUCTING AND ERECTING A GABLED ROOF

10.4.1.1 Laying out common rafters

10.4.1.2 Erecting gable roof frame

10.4.1.3 Setting out and fixing purlins

10.4.1.4 Selecting and applying roof coverings

10.4.1.5 Completing ridge coverings

UNIT A10.5 CONSTRUCTING AND ERECTING A FLAT ROOF

10.5.1 Types of flat roofs

10.5.2 Selecting building materials used in roof construction

10.5.3 Erection of flat and lean-to roofs as used in light construction

MODULE 70B2-11-A TEMPORAL TIMBER STRUCTURES

NOMINAL DURATION: 40 HOURS 4 CREDITS

PURPOSE: To equip the trainees with knowledge, skills and attitudes to enable them erect temporally timber structures in the construction industry.

OBJECTIVES: On completion of this module, trainees will be able to:

1. Erect centers for bricks and stone arches
2. Erect scaffolding and ladders

UNIT A11.1 ERECTING CENTRES FOR ARCHES

11.1.1 Types of centers
11.2.2 Purpose of centers
11.2.3 Setting out of centers
11.2.4 Support of centers
11.2.5 Easing and striking centers

UNIT A11.2 ERECTING SCAFFOLDING AND LADDERS

11.2.1 Types of scaffolding and ladders
11.2.2 Prepare the ground for scaffolding and ladders
11.2.3 Assemble parts in positions
11.2.4 Lay out the platforms
11.2.5 Dismantling scaffolds
11.2.6 Cleaning and storing